



Cyclistic Bike Share Analysis: A Deep Dive into User Patterns

This presentation explores the **usage patterns of Cyclistic's** bike sharing system, **comparing annual members and casual riders**. We aim to uncover key insights to guide marketing strategies and drive future growth.



Understanding User Patterns: Annual Members vs. Casual Users



Problem

Cyclistic offers various membership options, but they seek to **understand the distinct usage patterns** of annual members and casual users.



Objective

This analysis aims to examine **usage patterns by user type** to inform effective marketing strategies for **converting casual users into members**.



Data Description: A Deep Dive into Our Dataset

1 Data Period

The dataset spans **October 2023 to October 2024**, encompassing a full year of Cyclistic bike usage.

3 Data Consistency

Data quality was ensured through thorough **checks for duplicates and missing values**, resulting in a **reliable analysis**.

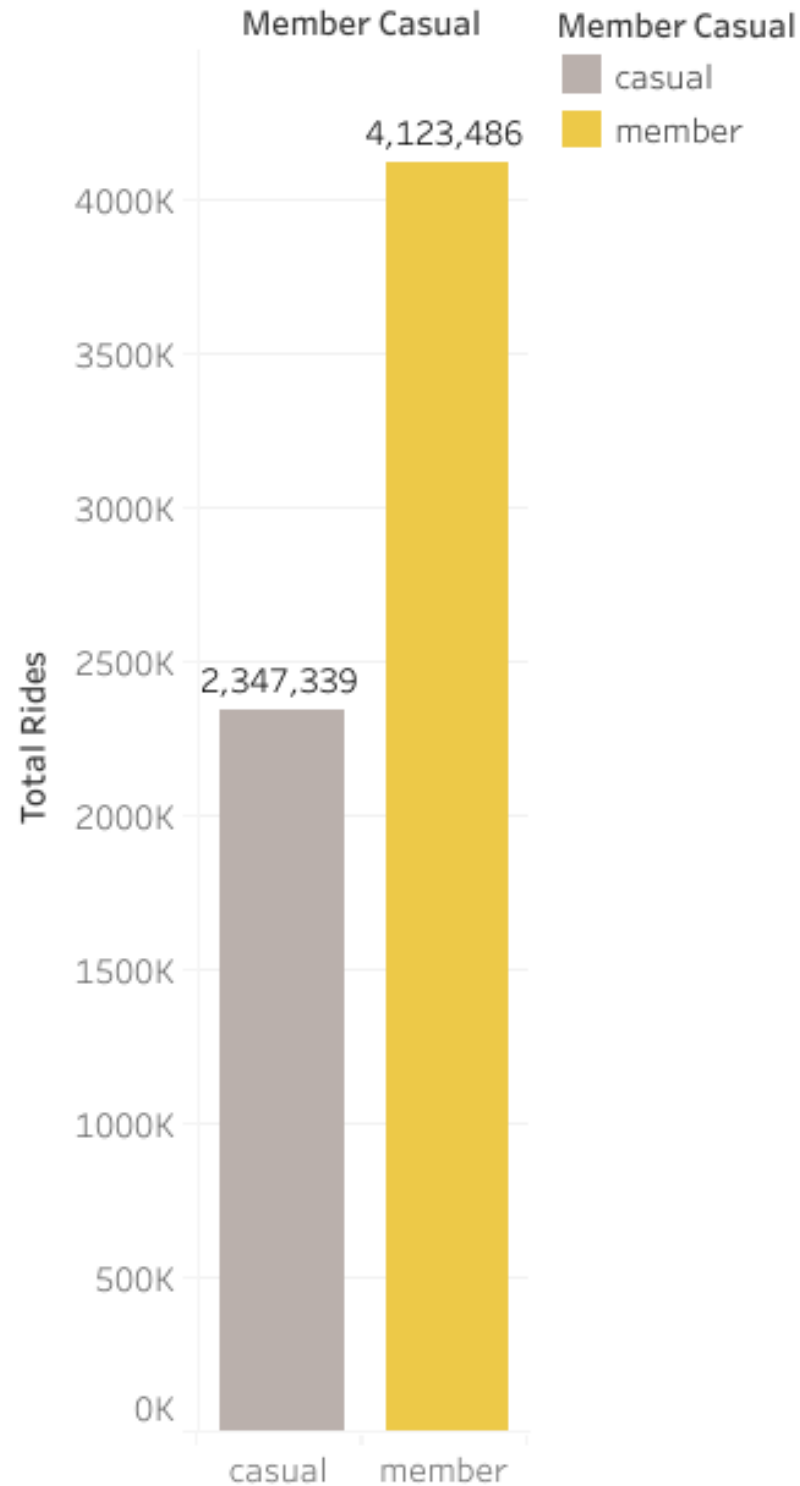
2 Data Mass

The dataset is comprehensive, including **information on user type, station locations, trip durations**, and other vital details.

4 Tools Used

This analysis leveraged powerful tools such as **R, BigQuery, and Tableau** for data manipulation, analysis, and visualization.

Total Rides by User Type



Usage per User Type: Unveiling the Patterns

4.1M

Member

2.3M

Casual

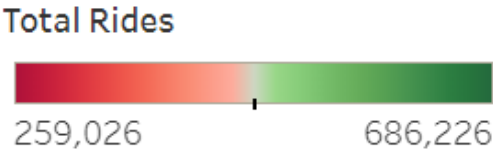
Annual members have significantly more rides than casual users, indicating a higher level of engagement with the service.

The majority of users are members, **representing approximately 64% of the total.** This **indicates good user retention** and successful **loyalty strategies.**

Daily Usage Patterns: Understanding Weekly Trends

Weekly Ride Distribution by User Type

Member Casual	Sunday	Monday	Tuesday	Wednes..	Thursday	Friday	Saturday
casual	472,854	406,048	275,869	259,026	298,789	297,053	337,700
member	520,504	463,501	589,121	637,819	686,226	653,367	572,948



Weekday Usage

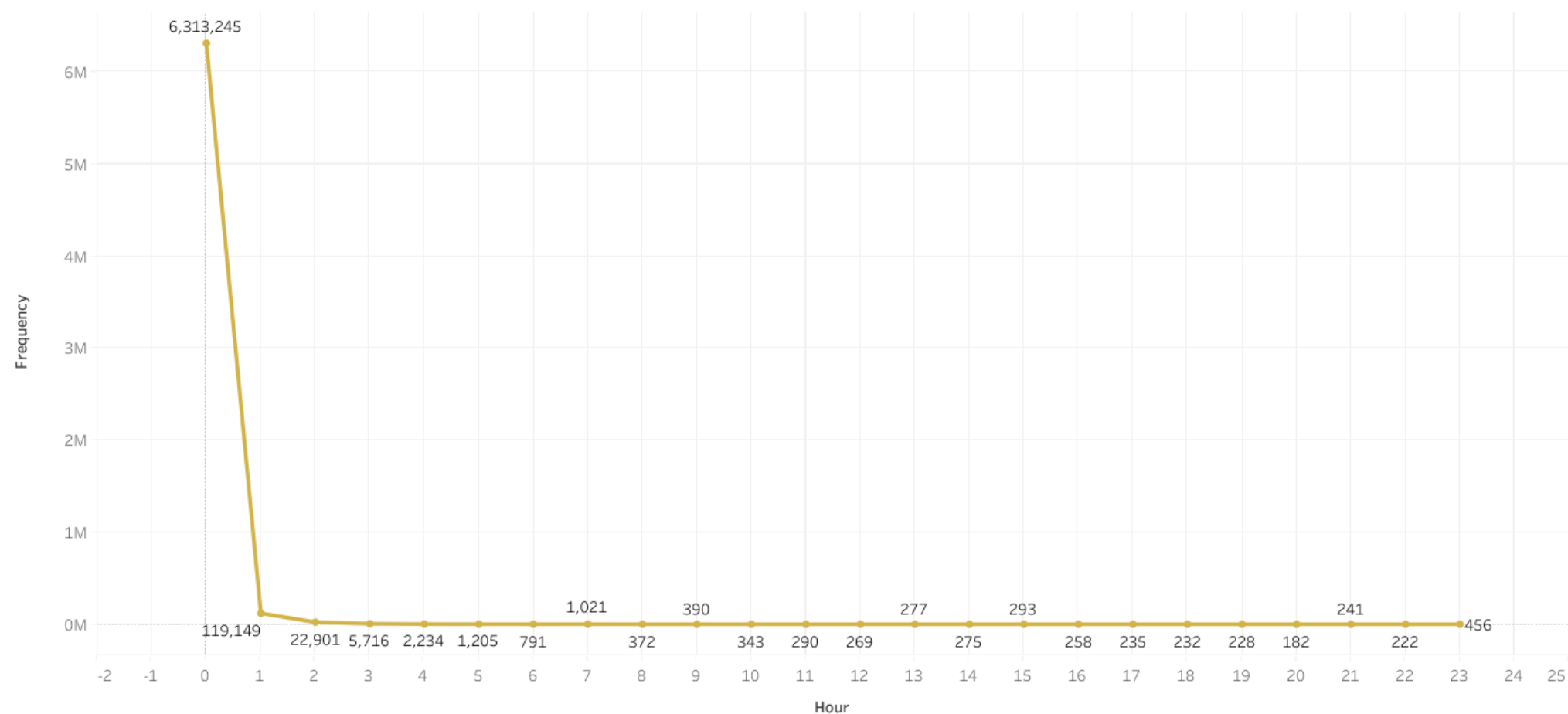
Both member and casual users exhibit distinct patterns. **Members** have consistent usage **throughout the week**, while **casual riders peak on weekends**.

Peak Ride Days

The total number of **rides is highest on Sunday (993,358)**, followed by **Thursday (985,015)**. This may indicate that the **service is popular both for leisure activities on weekends and for commuting during the week**.

Frequency of Rides by Hour

Frequency of Rides by Hour



The highest frequency of rides occurs at midnight (hour 0) with 6,313,245 rides, likely due to system resets or end-of-day processing. Ride frequency increases from **hour 7, indicating significant morning commute usage.** The **lowest frequency is between hours 4 and 6, with the lowest at hour 5 with 1,205 rides.** Understanding this pattern **helps optimize service availability and improve user experience.**

Max of ride length in hours
(member and casual)

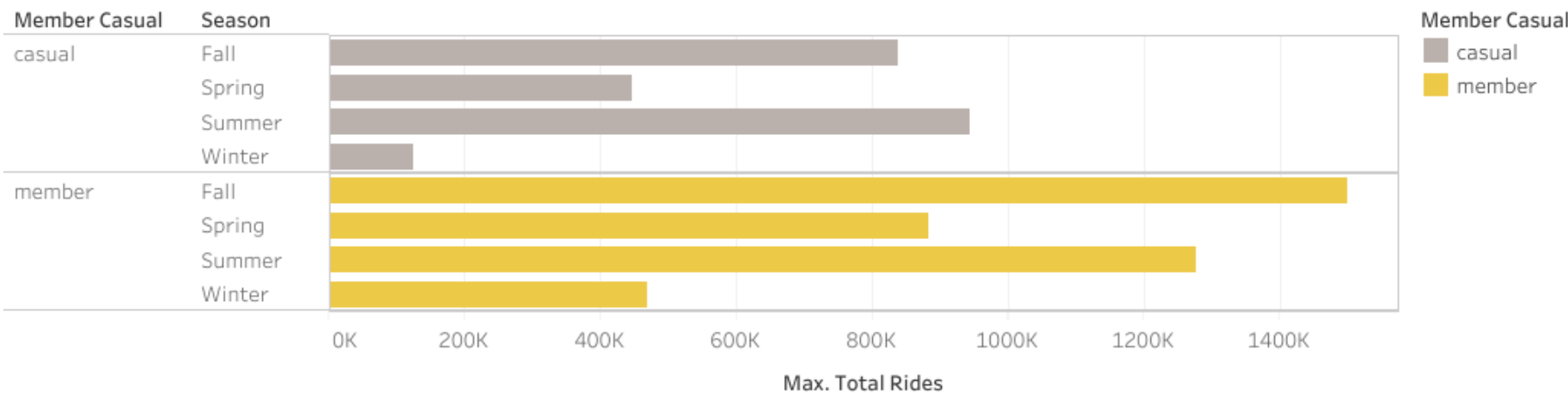
Member Casual	Ride Id	
casual	2E7A26B0B6513B10	25.999
	4F01365218A0D0C3	25.995
	5BC702AC59561D1F	25.994
	5C466397FA10F4E5	25.998
	7E79B2B0FAF61D31	25.995
	88D00324B450377C	25.998
	446B8A4213AAD837	25.999
	8631F8636227A814	25.994
	657208D4C95CF337	25.995
	DCEBA4BEFD05CBCC	25.996
member	1D78B93593DAC700	24.999
	32B0CE9A4BCDF4C0	24.999
	44C5CF5A018D4A1E	24.999
	53471F72F6F23E5D	24.999
	9901966A38BC90F0	24.999
	782523992134660D	25.008
	A243363027749E4B	25.997
	B289146098D45A62	24.999
	D740FF44BE912031	25.994
	ECB3DE14D3FBF698	24.999

Maximum Ride Length Analysis

The **data suggests unusually long ride times** that likely **indicate errors or anomalies**. It's important to **validate and clean the data** to **ensure accurate analysis and insights**.

User Behavior During Different Seasons (Members vs Casual Users)

User Behavior Changes During Different Seasons (Members vs Casual Users)



The highest number of rides for both groups occurs in summer, indicating a peak in usage during this season. **Members have significantly more rides in all seasons**, suggesting a **strong loyalty and regular usage pattern** among them. **Casual users** also show **increased activity in summer and fall**, likely due to favorable weather conditions.

Daily Usage Patterns: Understanding Weekly Trends

Most used start stations

Start Station Name	
Unknown	1,163,682
Streeter Dr & Grand Ave	70,331
DuSable Lake Shore Dr & ..	47,293
Kingsbury St & Kinzie St	42,777
Michigan Ave & Oak St	42,369
DuSable Lake Shore Dr & ..	41,778
Clark St & Elm St	38,978
Clinton St & Washington ..	38,586
Clinton St & Madison St	36,007
Millennium Park	35,661
Wells St & Concord Ln	35,383

Most used end stations

End Station Name	
Unknown	1,201,309
Streeter Dr & Grand Ave	71,561
DuSable Lake Shore Dr & ..	45,803
DuSable Lake Shore Dr & ..	45,170
Michigan Ave & Oak St	42,336
Kingsbury St & Kinzie St	42,002
Clinton St & Washington ..	39,266
Clark St & Elm St	38,342
Clinton St & Madison St	36,922
Millennium Park	35,960
Wells St & Concord Ln	35,456

High Usage

The "Unknown" station has an exceptionally high number of rides, which may indicate data recording issues or a large number of unregistered start locations.

Popular Stations

The most popular stations are located in central and high-traffic areas, indicating their importance for commuting and leisure activities.



Key Findings and Recommendations:

Popular Stations: The most popular stations are located in **central, high-traffic areas**, highlighting their importance for commuting and leisure activities.

Annual Members' Engagement: Annual members take **significantly more trips than casual users**, indicating a **higher level of engagement** with the service.

User Retention: Members constitute approximately 64% of the total users, reflecting successful **user retention and loyalty strategies**.

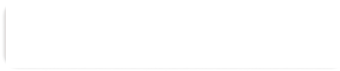
Usage Patterns: Members show consistent usage **throughout the week**, while **casual** users have peak usage **on weekends**.

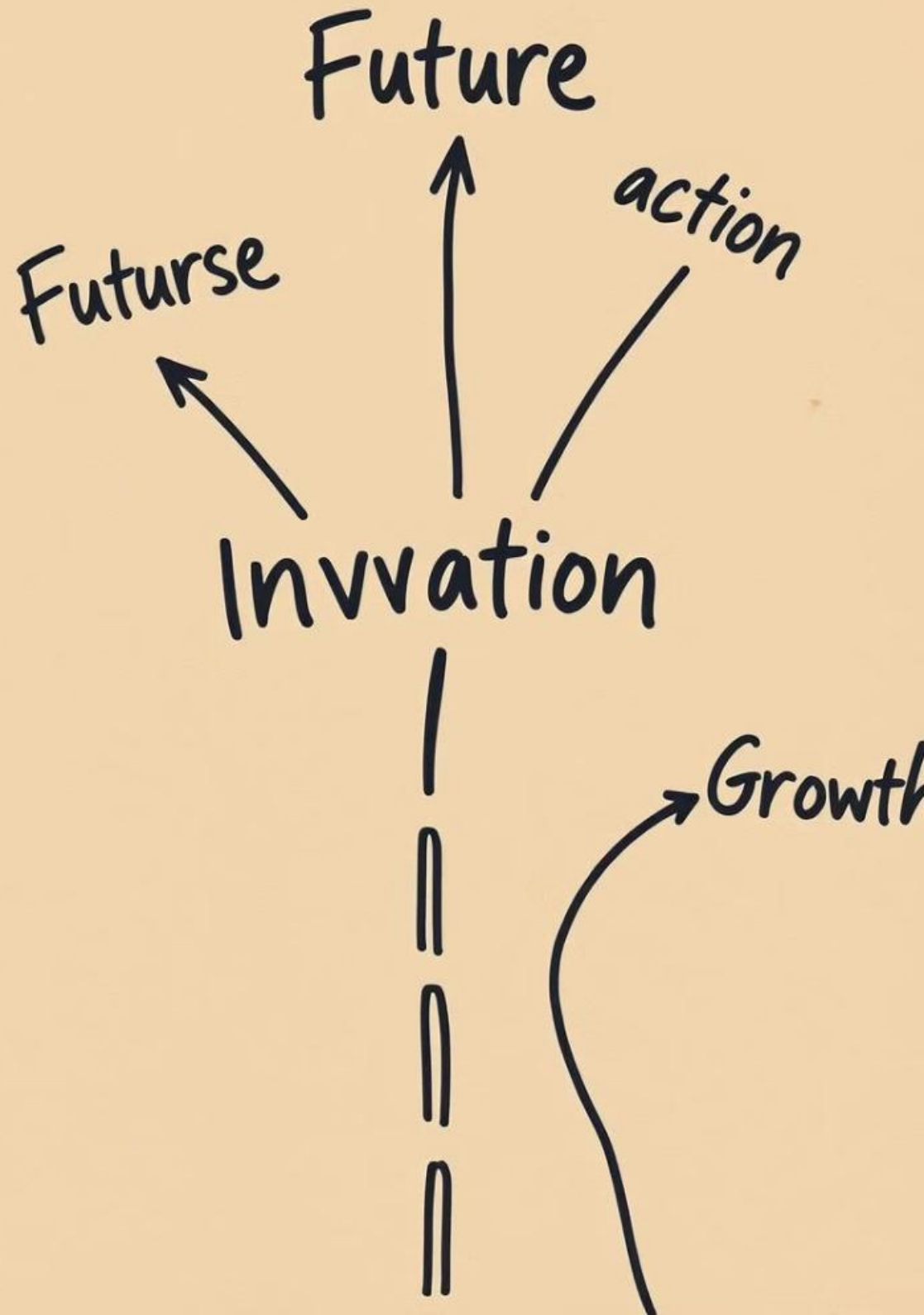


Recommendations for the Marketing Team:

Convert Casual Users to Annual Members: Encourage casual users to become annual members by **highlighting the benefits of consistent use and loyalty advantages.**

Weekday Usage Campaigns: Focus on marketing campaigns that promote **weekday usage for casual users, potentially converting them into annual members.**





Suggestions for Future Analysis and Data Collection Improvements

Conduct a deeper analysis of user demographics and trip characteristics to gain more detailed insights.

Enhance data collection to include user feedback on their experiences and preferences.

Develop targeted campaigns to increase weekday usage among casual users.

Create loyalty programs that encourage casual users to transition to annual memberships by highlighting long-term benefits.